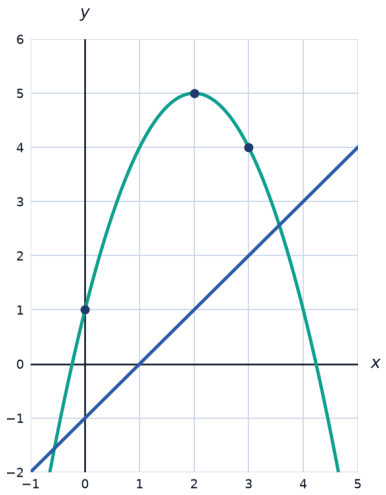
Geometric sequences as exponential functions

Core Concepts: Geometric sequences as exponential functions

1 In $\triangle ABC$ studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 105°

b. 115°

c. 97°

d. 150°

2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 10

c. 8

d. 12

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 98°

b. 108°

c. 90°

d. 150°

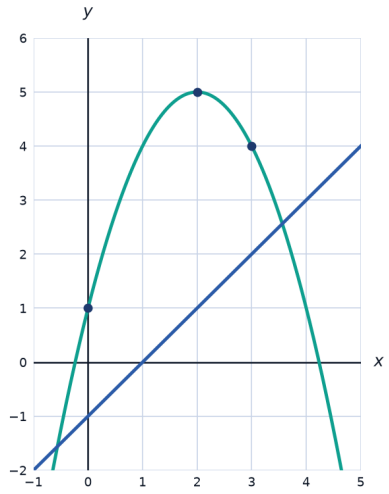
5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm

c. 16 cm

d. 36 cm



6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 8

b. 14

c. 12

d. 16

Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 In $\triangle ABC$ studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 91°

b. 101°

c. 83°

d. 150°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

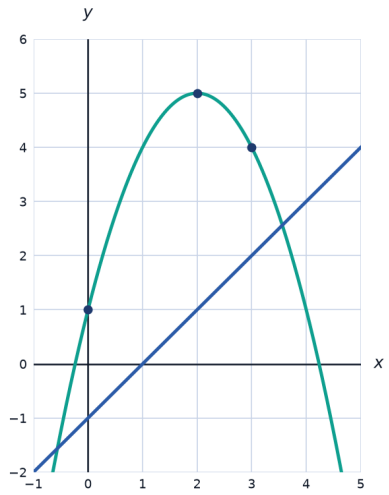
9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 12

b. 18

c. 16

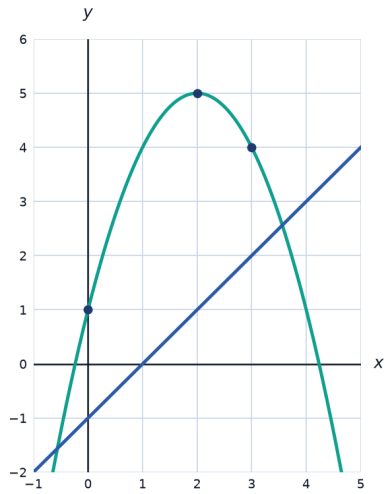
d. 20



Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In $\triangle ABC$ studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 84°

b. 94°

c. 76°

d. 150°

11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm

c. 24 cm

d. 100 cm

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 16

b. 22

c. 20

d. 24

© ClipSAT: By Mr. Mohamed Abdallah +966597688647